

Amir Karimi

Hartford, CT

karimiuconn@gmail.com

Summary of Key Attributes

Ph.D. candidate seeking a 3–12 month engineering internship starting in August. Relocation-ready and highly proficient in directing end-to-end projects from design through validation. A detailed professional portfolio is available on my GitHub [personal website](#).

Work Experience

UConn Neuroprosthesis Lab (Dr. Martin Han)

Lead Embedded Systems Engineer

August 2022 – May 2024

- Developed low-power neurostimulator circuits and bare-metal C firmware, integrating SPI/I2C/UART interfaces with hands-on oscilloscope and logic analyzer validation.

UConn Wearable Device Lab (Dr. Hugo Posada-Q)

PhD Candidate Researcher & Lead Hardware Engineer

May 2024 – July 2026

- Designed low-noise analog front-ends, managed KiCad multi-layer PCB layouts, and executed SPICE simulations alongside MATLAB system validation.

Technical Skills

Embedded Firmware Development: Bare-Metal C, STM32 (ARM Cortex-M), Low-Level Driver Development, Memory Mapping & Optimization

Hardware-Firmware Integration: Peripheral Interfacing (SPI, I2C, UART), Interrupt Handling (NVIC), DMA Configuration, Hardware Abstract Layers (HAL)

Signal Processing & Data Analysis: Real-Time Digital Signal Processing (DSP), Digital Filtering (FIR/IIR), Feature Extraction, MATLAB, Python

Hardware-Aware Design: Analog Front-End (AFE) Interfacing, Circuit Design & Schematic Capture, PCB Design (Altium & KiCad, Multi-Layer Layout), Power Supply Decoupling Architecture

Major Projects

Neurostimulator Front-End (STM32): Designed from scratch: custom stimulation/sensing circuit, multi-layer PCB, STM32 firmware, coherent demodulation, and a MATLAB control/DSP app.

Multimodal Wireless Biosignal Recorder (BLE): Built end-to-end: analog AFE with active filtering, BLE streaming, LTspice-validated design, SolidWorks enclosure, and a DSP + validation pipeline.

Education

Ph.D., Biomedical Engineering (*Hardware-based dissertation*)

Aug 2022 – Present

University of Connecticut

B.Sc., Biomedical Engineering (*Circuit and Instrumentation Track*)

2018 – 2022

Tehran Polytechnic University

Selected First Author Publications

1. Circuit-level analysis of a novel ternary adder in ternary-enabled graphene nanoribbon FETs technology | [Engineering Research Express](#)
2. Simulation-Based Evaluation of AC vs DC Electrodermal Activity Measurement Circuits for Long-Term Wearable Applications | [IEEE-EMBS International Conference on Body Sensor Networks](#)
3. Development and Validation of a Multimodal Wearable Belt for Abdominal Biosignal Monitoring with Application to Irritable Bowel Syndrome | [Micromachines](#)
4. Hybrid Cross-Coupled Inverter-Based 8T SRAM Design With Half-Selection Immunity and Enhanced Stability for Power-Constrained Applications | [IEEE Access](#)